9.4 Circle Equations

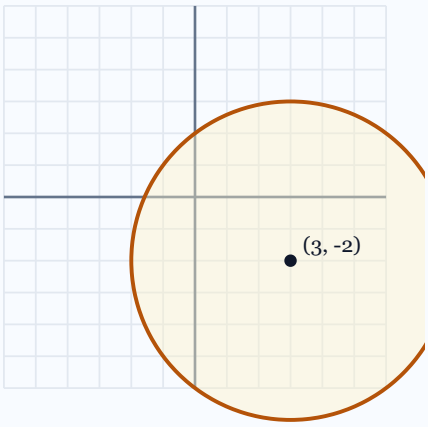
A circle is the set of points a fixed distance r from a center (h, k). Applying the distance formula to a general point (x, y) on the circle gives $\sqrt{(x-h)^2+(y-k)^2} = r$, and squaring both sides produces the **standard form**: $(x-h)^2+(y-k)^2 = r^2$. If an equation is given in expanded form ($x^2+y^2+Dx+Ey+F=0$), **completing the square** on the x-terms and the y-terms separately converts it back to standard form, revealing the center and radius. Once you have standard form, you can test whether a point is inside, on, or outside the circle by comparing $(x-h)^2+(y-k)^2$ to r^2: less than means inside, equal means on, greater means outside. This worksheet also finds where a circle crosses the axes and includes a coverage-radius application.

Examples

Example 1: Write the equation from center and radius

Write the equation of the circle with center $(3, -2)$ and radius 5.

Solution



Center $(3, -2)$, radius 5.

$(x-h)^2+(y-k)^2 = r^2$ with $h = 3, k = -2, r = 5$:

$(x-3)^2+(y+2)^2 = 25.$

Answer: $(x-3)^2+(y+2)^2 = 25.$

Example 2: Read the center and radius

Find the center and radius of $(x+1)^2+(y-4)^2=49$.

Solution

Identify the pattern $(x-h)^2+(y-k)^2=r^2$ hiding in the equation.
Rewrite $(x+1)^2$ as $(x-(-1))^2$ and 49 as 7^2 to match the pattern exactly:
$(x-(-1))^2+(y-4)^2=7^2$.
Reading off h, k, r gives the center and radius.
Answer: center $(-1, 4)$, radius 7.

Example 3: Center and a point on the circle

Write the equation of the circle centered at the origin passing through $(6, 8)$.

Solution

The radius equals the distance from the center to the point: $r=\sqrt{6^2+8^2}=\sqrt{36+64}=\sqrt{100}=10$.
$x^2+y^2=10^2=100$.
Answer: $x^2+y^2=100$.

Example 4: Complete the square

Rewrite $x^2+y^2-6x+4y-12=0$ in standard form. Find the center and radius.

Solution

Group and complete the square for each variable:
$(x^2-6x)+(y^2+4y)=12$
$(x^2-6x+9)+(y^2+4y+4)=12+9+4$
$(x-3)^2+(y+2)^2=25$.
Answer: center $(3, -2)$, radius 5.

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Example 5: Inside, on, or outside?

Is the point $(2, 2)$ inside, on, or outside the circle $(x-1)^2 + (y-1)^2 = 4$?

Solution

Substitute the point: $(2-1)^2 + (2-1)^2 = 1 + 1 = 2$.

Compare to $r^2 = 4$: since $2 < 4$, the point is **inside** the circle.

Example 6: Write the equation

Write the equation of the circle with center $(-4, 5)$ and radius 3.

Solution

Identify $h = -4$, $k = 5$, $r = 3$.

Substitute into $(x-h)^2 + (y-k)^2 = r^2$:

$(x-(-4))^2 + (y-5)^2 = 3^2$

$(x+4)^2 + (y-5)^2 = 9$.

Answer: $(x+4)^2 + (y-5)^2 = 9$.

Example 7: Read the equation

Find the center and radius of $(x-6)^2 + (y+2)^2 = 81$.

Solution

Identify the pattern $(x-h)^2 + (y-k)^2 = r^2$: here $h = 6$, $k = -2$, and $r^2 = 81$.

Take the square root to find the radius: $r = \sqrt{81} = 9$.

Answer: center $(6, -2)$, radius 9.

Example 8: Through a point

An origin-centered circle passes through $(-5, 12)$. Write its equation.

Solution

Identify that the radius equals the distance from the center $(0, 0)$ to the given point $(-5, 12)$.

$r = \sqrt{(-5)^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13$.

Substitute $r^2 = 169$ into $x^2 + y^2 = r^2$:

$x^2 + y^2 = 169$.

Answer: $x^2 + y^2 = 169$.

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Example 9: Complete the square (both variables)

Rewrite $x^2 + y^2 + 8x - 2y + 8 = 0$ in standard form. Find the center and radius.

Solution

$(x^2 + 8x) + (y^2 - 2y) = -8$

$(x^2 + 8x + 16) + (y^2 - 2y + 1) = -8 + 16 + 1$

$(x + 4)^2 + (y - 1)^2 = 9.$

Answer: center $(-4, 1)$, radius 3.

Example 10: Where a circle crosses the x-axis

Find the points where the circle $(x - 2)^2 + (y - 3)^2 = 25$ crosses the x-axis.

Solution

On the x-axis, $y = 0$. Substitute:

$(x - 2)^2 + (0 - 3)^2 = 25$

$(x - 2)^2 + 9 = 25$

$(x - 2)^2 = 16$

$x - 2 = \pm 4 \Rightarrow x = 6$ or $x = -2$.

Answer: $(6, 0)$ and $(-2, 0)$.

Example 11: Application — cell tower coverage

A cell tower at the origin of a map (in km) has a coverage circle of radius 8 km: $x^2 + y^2 = 64$. A house is located at $(5, 7)$. Is the house within range?

Solution

Compute the squared distance from the tower to the house: $5^2 + 7^2 = 25 + 49 = 74$.

Compare to $r^2 = 64$: since $74 > 64$, the house is **outside** the coverage circle.

(Actual distance = $\sqrt{74} \approx 8.60$ km, slightly beyond the 8 km range.)

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Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Write equation

Center $(2, -5)$, radius 4. Write the equation.

Answer: **Verification:**

Problem 2: Read equation

Find center and radius of $(x + 3)^2 + (y - 7)^2 = 36$.

Answer: **Verification:**

Problem 3: Through a point

Center $(0, 0)$ through $(3, 4)$. Write the equation.

Answer: **Verification:**

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Problem 4: Complete the square

Rewrite $x^2 + y^2 - 4x + 6y - 3 = 0$ in standard form.

Answer: **Verification:**

Problem 5: Point test

Is $(5, 3)$ inside, on, or outside $(x - 2)^2 + (y + 1)^2 = 36$?

Answer: **Verification:**

Problem 6: Write equation

Center $(-1, -1)$, radius 2. Write the equation.

Answer: **Verification:**

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Problem 7: Read equation

Find center and radius of $x^2 + (y - 3)^2 = 1$.

Answer: **Verification:**

Problem 8: Through a point

Center $(1, 2)$ through $(4, 6)$. Write the equation.

Answer: **Verification:**

Problem 9: Complete the square

Rewrite $x^2 + y^2 + 2x - 10y + 17 = 0$ in standard form.

Answer: **Verification:**

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Problem 10: Point test

Is $(0, 0)$ on $x^2 + y^2 = 1$?

Answer: **Verification:**

Problem 11: Radius from equation

What is r for $(x - 5)^2 + (y + 4)^2 = 64$?

Answer: **Verification:**

Problem 12: Center from equation

What is the center of $(x + 9)^2 + y^2 = 4$?

Answer: **Verification:**

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Problem 13: Point test

Is $(3, 4)$ on $x^2 + y^2 = 25$?

Answer: **Verification:**

Problem 14: Challenge

Find where circle $(x - 4)^2 + (y - 1)^2 = 25$ crosses the y-axis.

Answer: **Verification:**

Problem 15: Challenge

Rewrite $x^2 + y^2 - 10x - 4y + 13 = 0$ in standard form, then determine whether $(7, 6)$ is inside, on, or outside the circle.

Answer: **Verification:**

✓ Answer Key

1. $(x-2)^2+(y+5)^2=16$
2. Center $(-3, 7)$, radius 6
3. $x^2+y^2=25$
4. $(x-2)^2+(y+3)^2=16$
5. Inside $(25<36)$
6. $(x+1)^2+(y+1)^2=4$
7. Center $(0, 3)$, radius 1
8. $(x-1)^2+(y-2)^2=25$
9. $(x+1)^2+(y-5)^2=9$
10. No — inside $(0<1)$
11. 8
12. $(-9, 0)$
13. Yes (on the circle)
14. $(0, 4)$ and $(0, -2)$
15. $(x-5)^2+(y-2)^2=16$; $(7, 6)$ is outside $(20>16)$

Great work — keep practising!